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CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, CAD/CAM)

May 2013

ME – 701.01 Computer Aided Design

Date: 11.05.2013, Saturday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator & PSG data book is allowed.

SECTION- I

Q1	(a)	How does the CAD help to reduce product development cycle? Explain.	4
	(b)	State the modelling facilities that are expected in good solid modelling software.	4
	(c)	By giving suitable examples explain the various techniques used for solid modeling.	5
Q2	(a)	Compare the capabilities of types of displays.	3
	(b)	What is scan conversion? Explain scan conversion of a line.	3
	(c)	A line joining point P (12, 12) and Q (20, 18) is to be drawn using Bresenham's algorithm. Find the coordinates of pixels starting from point P.	5
		OR	
Q2	(a)	Differentiate between: 1. Analytic curves and Parametric curves 2. Passive and Interactive graphics	3
	(b)	Explain: "Bresenham's algorithm is more efficient than the DDA line generation algorithm".	3
	(c)	Write Bresenham's circle drawing algorithm with (x_c, y_c) as center and radius r.	5
Q3	(a)	What is homogenous coordinate system? Why is it preferred in computer graphics?	3
	(b)	Compare geometric transformation with coordinate transformation. Also explain composite transformation.	3
	(c)	A triangle PQR has its vertices at P (4, 2), Q (8, 2) & R (6, 5). It is to be rotated in CCW direction about the point (6, 5) through 90°. Find the new position of the triangle.	5
		OR	
Q3	(a)	Write the matrices for the following 2-D transformations: (i) Scaling (ii) Rotation (iii) Shearing.	3
	(b)	Obtain 3D rotations for cases: (i) Rotation about x-axis followed by equal rotation about y-axis. (ii) Rotation about x-axis followed by equal rotation about y-axis. Hence prove that 3D rotations are not commutative.	3
	(c)	Show that the transformation matrix for reflection about the line Y = - X is equivalent to reflection relative to Y axis followed by counterclockwise rotation by 90°.	5

SECTION- II

Q4 (a) Write down the steps for generation of solid model of a Hexagonal Nut using any solid modeling software. 4

(b) Explain parametric continuity conditions for curves. 4

(c) Explain feature based models. How are those constructed? 4

Q5 (a) Explain rotary mode and plane mode of scanning techniques & list out different reverse engineering software's. 5

(b) The coordinates of four control points relative to a current WCS are given by $P_0 = [2\ 2\ 0]^T$, $P_1 = [2\ 3\ 0]^T$, $P_2 = [3\ 3\ 0]^T$ & $P_3 = [3\ 2\ 0]^T$. Find the equation of the resulting Bezier curve. Also find points on curve for u = 0, ¼, ½, ¾, & 1. 6

OR

Q5 (a) What is reverse engineering? Justify its use. 3

(b) Explain various properties of Bezier curves. 4

(c) Calculate the parametric midpoint of Hermite cubic curve that fits the points P_0 = (1, 3), P_1 = (7, 2) and the tangent vectors P'_0 = (10, 8), P'_1 = (6, 0). 4

Q6 (a) Write a short notes on the following: 6

(i) DXF (ii) NURBS

(b) For design of shaft prepare an interactive programme using any programming language. 6

OR

Q6 (a) Discuss the need for standardization in computer graphics & list the various CAD standards. 6

(b) Prepare an exhaustive algorithm for design of connecting rod of an IC engine 6

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